QUANT

Permutation, Combination & Probability

Assignment 02

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P & C Assignments 02

1. The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of three parallel lines is

1. 6 2. 9 3. 12 4. 18

2. The number of ways in which a mixed double game can be arranged from amongst 9 married couples if no husband and wife play in the same game is

1. 756 2. 1,512 3. 3,024 4. None of these

3. Between two junctions A and B there are 12 intermediate stations. The number of ways in which a train can be made to stop at 4 of these stations so that no two of the halting stations are consecutive is

1. ${}^{8}C_4$ 2. ${}^{9}C_4$ 3. ${}^{12}C_4 - 4$ 4. None of these

4. Eight chairs are numbered 1 to 8. Two women and three men wish to occupy one chair each. First the women chose the chairs from amongst the chair marked 1 to 4; and then the men select the chairs from amongst the remaining. The number of possible arrangements is

1. ${}^{4}C_3 \times {}^{4}C_2$ 2. ${}^{4}C_2 \times {}^{4}P_3$ 3. ${}^{4}P_2 \times {}^{4}P_3$ 4. None of these

5. There are 4 balls of different colours and four boxes of colours, same as those of those of the balls. The number of ways in which the balls, one in each box, could be placed such that a ball does not go to box of its own colour is

1. 9 2. 24 3. 12 4. None of these

6. From a class of 12 boys and 10 girls, 10 students are to be chosen for a competition, including at least 4 boys and 4 girls. The 2 girls who won the prizes last year have to be included. In how many ways can such a selection be made?

1. ${}^{8}C_4 \times {}^{12}C_4 + {}^{8}C_3 \times {}^{12}C_5 + {}^{8}C_2 \times {}^{12}C_6$ 2. ${}^{8}C_4 \times {}^{12}C_4 + {}^{8}C_3 \times {}^{12}C_5$
3. ${}^{8}C_3 \times {}^{12}C_5 + {}^{8}C_2 \times {}^{12}C_6$ 4. ${}^{8}C_4 \times {}^{12}C_4 + {}^{8}C_3 \times {}^{12}C_5 + {}^{8}C_2 \times {}^{12}C_5$

7. Everybody in a room shakes hands with everybody else. The total number of handshakes is 66. The total number of persons in the room is -

1. 10 2. 11 3. 12 4. 13

8. A candidate is required to answer 7 questions out to 12 questions, which are divided into two groups each containing 6 questions. He is not permitted to attempt more than 5 questions from each group. In how many ways can he choose the 7 questions?

1. ${}^{12}C_7$ 2. ${}^{12}C_7 - 2 \times {}^{6}C_1$ 3. ${}^{12}C_7 - 2 \times 6!$ 4. None of these

9. The odds that A speaks the truth are 3 : 2 and the odds that B speaks the truth are 5 : 3. In what percentage of cases are they likely to contradict each other on an identical point?

1. 40% 2. 25% 3. 50% 4. 47.5%

10. Two dice are thrown. Find the probability that a multiple of 2 occurs on one dice and a multiple of 3 occurs on the other.

1. $\frac{10}{36}$ 2. $\frac{9}{36}$ 3. $\frac{11}{36}$ 4. $\frac{12}{36}$

Answer key & Explanations

P & C Assignments 02

1.	4	Since each parallelogram is formed by choosing two parallel straight lines from the first and two from the second set. $\therefore$ total no. of parallelograms formed = ${}^4C_2 \times {}^3C_2 = 6\times3=18$
2.	2	Out of 9 men two men can be chosen in ${}^9C_2$ ways. Since no husband and wife are to play in the same game, so we have to select two women from the remaining 7 women. This can be done in ${}^7C_2$ ways. If M_1M_2 and W_1W_2 are chosen then a team can be constituted in 4 ways viz $M_1W_2;M_1W_1;M_2W_1;M_2W_2$ Thus the number of ways of arranging the game is $={}^9C_2\times{}^7C_2\times2=1512$.
3.	2	$\leftarrow x_1 \rightarrow\leftarrow x_2 \rightarrow\leftarrow x_3 \rightarrow\leftarrow x_4 \rightarrow\leftarrow x_5 \rightarrow$ $A \quad 1 \quad 2 \quad 3 \quad 4 \quad B$ Here $x_1+x_2+x_3+x_4+x_5=8(1)$ $x_1, x_{2,....} x_5$ is the no. of intermediate stns. (1, 2, 3, 4) are the stations. Here $x_1\geq0;x_5\geq0,x_2,x_3,x_4\geq1$ The total number of ways is the number of solutions of the above equation. Let $y_2=x_2-1;y_3=x_3-1;y_4=x_4-1;$ There for equation (1) reduces $x_1+y_2+y_3+y_4+x_5=5$ Where $y_2.y_3.y_4\geq0;$ The no. of solution $={}^{5+5-1}C_{5-1}={}^9C_4$
4.	4	The required no.$={}^4P_2\times{}^6P_3$
5.	1	The required no. of ways = $4! - 1 - {}^4C_2 - 2\,({}^4C_3) = 9$
6.	1	Since 2 girls are already chosen only 8 students are to be chosen. The different ways in which the selection of the 8 people can be made are as follows. 4 girls and 4 boys can be chosen in ${}^8C_4\times{}^{12}C_4$ ways. 3 girls and 5 boys can be chosen in ${}^8C_3\times{}^{12}C_5$. 2 girls and 6 boys can be chosen in ${}^8C_2\times{}^{12}C_6$. Therefore the total number of ways in which the selection can be done $={}^8C_4\times{}^{12}C_4+{}^8C_3\times{}^{12}C_5+{}^8C_2\times{}^{12}C_6$.
7.	2	If there are $n+1$ n men in the room then the first man shakes hands n times and the second man $n-1$ times and the third man $n-2$ times. Therefore the sum of the shake hands $=n+(n-1)+(n-2)+...+1=n(n+1)/2=66$. Therefore n = 11. Hence the number of men is 11.
8.	2	The candidate has to answers 7 questions out of 12. The number of combinations of 7 questions out of $12={}^{12}C_7$. But these will contain combination in which the candidate has chosen 6 questions from one section and only one from the other section. This can be done in ${}^6C_6\times{}^6C_1$ ways. This can be done with the other section also so another ${}^6C_6\times{}^6C_1$ combinations become invalid. Therefore the number of ways of section of question $={}^{12}C_7-2\times{}^6C_1$.
9.	4	There are two cases where the two would surely contradict each other. The first case where A speaks the truth and B lies. The probability of such a case $=\frac{3}{5}\times\frac{3}{8}=\frac{9}{40}$ Here we have the product of the probabilities because two independent are to performed. The second case where A and B would contradict each other is when A lies and B speaks the truth. The probability of such an event $=\frac{2}{5}\times\frac{5}{8}=\frac{10}{40}$. The probabilities of the two independent cases $=\frac{9}{40}+\frac{10}{40}=\frac{19}{40}=0.475=47.5\%$
10.	3	We have a total of 11 favorable cases viz. 2,3 2,6, 4,3 4,6 6,3 3,6 6,4 3,4 6,2 3,2 and 6,6 out of a

		total of 36 cases. So the answer is $\frac{11}{36}$.